

Accuracy analyses

This vignette contains a full breakdown of the analyses of participants' accuracy on the reasoning problems. The most interesting results are synthetically presented in the main manuscript (preprint here). The present document aims to provide a full account of the steps taken to reach these results, and to provide additional analyses that were not included in the manuscript for brevity.

```
library(aphantasiaReasoningViie)
#> Welcome to aphantasiaReasoningViie.
```

Data preparation

First, let's get the clean, analysis-ready data, create cognitive style clusters using OSIVQ scores (see vignette("preparing_data") and vignette("osivq_clusters") for details), and add this classification to the dataset containing accuracy data.

```
df_survey <- get_clean_data()$df_survey
# Clustering OSIVQ data
clustering <- cluster_osivq(df_survey)
# Adding named clusters to the survey data
df_survey <- add_named_clusters(
  df_survey, clustering,
  names = c("Spatialiser", "Visualiser", "Verbaliser"),
  levels = c("Visualiser", "Spatialiser", "Verbaliser"),
  contrasts = c("_visualiser", "_spatialiser", "_verbaliser"),
  base = 1
)
# Merging with experiment data
df_expe <-
  dplyr::left_join(
    get_clean_data()$df_expe,
    df_survey |> dplyr::select(id, cluster),
    by = dplyr::join_by("id")
  ) |>
  dplyr::relocate(cluster, .after = "group") |>
  dplyr::select(id, group:strategy_group, problem, category, accuracy)

dplyr::glimpse(df_expe)
#> Rows: 2,808
#> Columns: 9
#> $ id          <fct> acdn247721443631359lzb, acdn247721443631359lzb, acdn2~
#> $ group       <fct> Typical, Typical, Typical, Typical, Typical, Typical, T~
#> $ cluster     <fct> Visualiser, Visualiser, Visualiser, Visualiser, Visuali~
#> $ group_2     <fct> Typical, Typical, Typical, Typical, Typical, Typical, T~
#> $ group_3     <fct> Typical, Typical, Typical, Typical, Typical, Typical, T~
#> $ strategy_group <fct> No visual strategy, No visual strategy, No visual strat~
#> $ problem     <int> 18, 25, 2, 19, 1, 10, 6, 9, 8, 26, 5, 15, 23, 14, 17, 2~
#> $ category    <fct> Spatial, Control, Visual, Control, Visual, Spatial, Vis~
```

```
#> $ accuracy      <int> 1, 1, 0, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
```

Method

As described in the manuscript:

We fitted Generalised Linear Mixed Models (GLMMs) with binomial distributions and logit links using the `glmmTMB` package (McGillucuddy et al., 2025) to predict accuracy with a grouping variable (VVIQ groups, OSIVQ clusters), Category (visual, spatial, or control) along with their two-way interactions as fixed categorical predictors. Varying slopes and intercepts (“random effects”) have been added for each participant by category and for each problem by grouping variable.

Let’s break this down.

The grouping variables

We used several grouping variables to classify participants, all of which are in the `df_expe` data frame:

- **group** is the 4-group VVIQ classification with an “aphantasia” group (VVIQ = 16), “hypophantasia” group (VVIQ ∈ [17, 32]), “typical” group (VVIQ ∈ [33, 74]), and “hyperphantasia” group (VVIQ ∈ [75, 80]). It was not used in the analyses because the hyperphantasia group was too small (N = 4).
- **group_3** is the same as **group**, but with the hyperphantasia group merged with the typical group. It is referred to as “VVIQ 3 groups” in the manuscript results.
- **group_2** is the binary classification often used in aphantasia literature that additionally merges the hypophantasia group with the aphantasia group. It is referred to as “VVIQ 2 groups” in the manuscript results.
- **cluster** is the cognitive style classification obtained from clustering OSIVQ scores. It contains three groups: “Visualiser”, “Spatialiser”, and “Verbaliser”. It is referred to as “OSIVQ 3 clusters” in the manuscript results.
- **strategy_group** is a classification based on the self-reported strategies used by the participants to solve the problems. It focuses on whether participants used a visual mental imagery strategy and contains two groups: “Visual strategy user” and “No visual strategy”. It is not reported in the manuscript because it was suggested to us after submission by colleagues at a conference.

The same modelling pipeline was therefore applied four times, once for each of the last four grouping variables.

The modelling pipeline

We created a small `build_formula()` helper function to write the formula easily, as we used the same model structure a lot of times:

```
build_formula("accuracy", "group_2")
#> accuracy ~ group_2 * category + (category | id) + (group_2 |
#>      problem)
#> <environment: 0x000001fe04b0a7b0>

build_formula("accuracy", "cluster")
#> accuracy ~ cluster * category + (category | id) + (cluster |
#>      problem)
#> <environment: 0x000001fe04ba6af8>
```

The models were fitted using the `glmmTMB::glmmTMB()` function (and a little `set_ranef_prior()` helper function to set a weakly informative prior on the random effects, which helps with convergence and singularity

issues). After model fit, we checked for potential singularity issues and model performance using the `get_singularity()` and `get_performance()` functions created for the occasion (which are convenient wrappers around the *performance* package).

Finally, we tested our hypotheses with marginal contrasts. As stated in the manuscript:

Due to the way that variance is partitioned in GLMMs (Rights & Sterba, 2019), there does not exist an agreed-upon way to calculate standard effect sizes for individual terms such as main effects or interactions in these models. Thus, in line with general recommendations on how to report effect sizes (e.g., Pek & Flora, 2018), we report and analyse unstandardised effect sizes for post-hoc tests in the form of estimated marginal contrasts (i.e. differences in model-estimated marginal means, hereinafter denoted Δ), in seconds for RTs or as odds ratios for accuracies. To answer our hypotheses, we planned to analyse contrasts between groups, contrasts between categories for each group separately, and interaction contrasts testing the differences in category contrasts between the groups.

This task was performed with the `report_contrast()` function, which is a wrapper around several functions from the *emmeans* package. We used it to compute accuracy contrasts between groups, contrasts between categories within each group, and interaction contrasts (differences in category contrasts between groups).

Here we go!

Results

VVIQ 2 groups

```
m_acc_vviq_2 <-
  glmmTMB::glmmTMB(
    data      = df_expe,
    formula   = build_formula("accuracy", "group_2"),
    family    = binomial(link = "logit"),
    prior     = set_ranef_prior(55)
  )

# Singularity
m_acc_vviq_2 |> get_singularity()
#> The model is not singular, parameter estimates are trustworthy.

# Performance
m_acc_vviq_2 |> get_performance() |> knitr::kable(align = "c")
```

AIC	BIC	R2 (cond.)	R2 (marg.)	ICC	RMSE
2915.940	3005.043	0.201	0.012	0.191	0.380

```
# Group contrasts
m_acc_vviq_2 |> report_contrast(~ group_2) |> knitr::kable()
#> NOTE: Results may be misleading due to involvement in interactions
```

Contrast	Odds ratio	95% CI	p.value
Aphantasia / Typical	0.829	[0.58, 1.18]	0.301

```
# Category contrasts within groups
```

```
m_acc_vviq_2 |> report_contrast(~ category | group_2) |> knitr::kable()
```

Contrast	group_2	Odds ratio	95% CI	p.value
Control / Spatial	Aphantasia	1.061	[0.45, 2.5]	0.986
Control / Visual	Aphantasia	1.081	[0.46, 2.57]	0.976
Spatial / Visual	Aphantasia	1.019	[0.43, 2.4]	0.999
Control / Spatial	Typical	1.335	[0.72, 2.48]	0.517
Control / Visual	Typical	1.906	[1.04, 3.5]	0.034
Spatial / Visual	Typical	1.427	[0.8, 2.55]	0.323

```
# Interaction contrasts
```

```
m_acc_vviq_2 |> report_contrast(~ category * group_2, interaction = TRUE) |>
  knitr::kable()
```

Category contrast	group_2_pairwise	Odds ratio	95% CI	p.value
Control / Spatial	Aphantasia / Typical	0.795	[0.45, 1.41]	0.430
Control / Visual	Aphantasia / Typical	0.567	[0.32, 1.01]	0.056
Spatial / Visual	Aphantasia / Typical	0.714	[0.41, 1.25]	0.238

VVIQ 3 groups

```
m_acc_vviq_3 <-
  glmmTMB::glmmTMB(
    data = df_expe,
    formula = build_formula("accuracy", "group_3"),
    family = binomial(link = "logit"),
    prior = set_ranef_prior(20)
  )
```

```
m_acc_vviq_3 |> get_singularity()
#> The model is not singular, parameter estimates are trustworthy.
```

```
m_acc_vviq_3 |> get_performance() |> knitr::kable(align = "c")
```

AIC	BIC	R2 (cond.)	R2 (marg.)	ICC	RMSE
2919.357	3044.102	0.208	0.013	0.198	0.379

```
m_acc_vviq_3 |> report_contrast(~ group_3) |> knitr::kable()
```

```
#> NOTE: Results may be misleading due to involvement in interactions
```

Contrast	Odds ratio	95% CI	p.value
Aphantasia / Hypophantasia	1.263	[0.68, 2.34]	0.650
Aphantasia / Typical	0.919	[0.55, 1.52]	0.918
Hypophantasia / Typical	0.727	[0.41, 1.28]	0.384

```
m_acc_vviq_3 |> report_contrast(~ category | group_3) |> knitr::kable()
```

Contrast	group_3	Odds ratio	95% CI	p.value
Control / Spatial	Aphantasia	0.971	[0.34, 2.75]	0.998
Control / Visual	Aphantasia	1.056	[0.37, 3.02]	0.992
Spatial / Visual	Aphantasia	1.088	[0.38, 3.08]	0.981
Control / Spatial	Hypophantasia	1.227	[0.48, 3.14]	0.866
Control / Visual	Hypophantasia	1.107	[0.42, 2.91]	0.967
Spatial / Visual	Hypophantasia	0.902	[0.35, 2.31]	0.964
Control / Spatial	Typical	1.333	[0.72, 2.45]	0.513
Control / Visual	Typical	1.901	[1.04, 3.46]	0.032
Spatial / Visual	Typical	1.426	[0.8, 2.53]	0.315

```
m_acc_vviq_3 |> report_contrast(~ category * group_3, interaction = TRUE) |>
  knitr::kable()
```

Category contrast	group_3_pairwise	Odds ratio	95% CI	p.value
Control / Spatial	Aphantasia / Hypophantasia	0.791	[0.37, 1.69]	0.546
Control / Visual	Aphantasia / Hypophantasia	0.954	[0.43, 2.11]	0.908
Spatial / Visual	Aphantasia / Hypophantasia	1.206	[0.56, 2.6]	0.632
Control / Spatial	Aphantasia / Typical	0.729	[0.36, 1.48]	0.380
Control / Visual	Aphantasia / Typical	0.556	[0.27, 1.14]	0.109
Spatial / Visual	Aphantasia / Typical	0.763	[0.38, 1.54]	0.449
Control / Spatial	Hypophantasia / Typical	0.921	[0.45, 1.88]	0.822
Control / Visual	Hypophantasia / Typical	0.582	[0.28, 1.22]	0.152
Spatial / Visual	Hypophantasia / Typical	0.632	[0.31, 1.28]	0.204

OSIVQ 3 clusters

```
m_acc_osivq <-
  glmmTMB::glmmTMB(
    data = df_expe,
    formula = build_formula("accuracy", "cluster"),
    family = binomial(link = "logit"),
    prior = set_ranef_prior(15)
  )

m_acc_osivq |> get_singularity()
#> The model is not singular, parameter estimates are trustworthy.

m_acc_osivq |> get_performance() |> knitr::kable(align = "c")
```

AIC	BIC	R2 (cond.)	R2 (marg.)	ICC	RMSE
2916.681	3041.426	0.203	0.010	0.195	0.380

```
m_acc_osivq |> report_contrast(~ cluster) |> knitr::kable()
#> NOTE: Results may be misleading due to involvement in interactions
```

Contrast	Odds ratio	95% CI	p.value
Visualiser / Spatialiser	1.023	[0.57, 1.83]	0.995
Visualiser / Verbaliser	1.179	[0.74, 1.87]	0.681
Spatialiser / Verbaliser	1.152	[0.65, 2.04]	0.830

```
m_acc_osivq |> report_contrast(~ category | cluster) |> knitr::kable()
```

Contrast	Cluster	Odds ratio	95% CI	p.value
Control / Spatial	Visualiser	1.380	[0.72, 2.66]	0.484
Control / Visual	Visualiser	1.853	[0.97, 3.55]	0.067
Spatial / Visual	Visualiser	1.343	[0.72, 2.5]	0.505
Control / Spatial	Spatialiser	0.997	[0.35, 2.81]	1.000
Control / Visual	Spatialiser	1.412	[0.5, 3.96]	0.713
Spatial / Visual	Spatialiser	1.417	[0.51, 3.93]	0.703
Control / Spatial	Verbaliser	1.115	[0.49, 2.54]	0.948
Control / Visual	Verbaliser	1.197	[0.52, 2.74]	0.867
Spatial / Visual	Verbaliser	1.073	[0.47, 2.44]	0.978

```
m_acc_osivq |> report_contrast(~ category * cluster, interaction = TRUE) |>
  knitr::kable()
```

Category contrast	Cluster contrast	Odds ratio	95% CI	p.value
Control / Spatial	Visualiser / Spatialiser	1.384	[0.64, 2.97]	0.404
Control / Visual	Visualiser / Spatialiser	1.312	[0.61, 2.83]	0.490
Spatial / Visual	Visualiser / Spatialiser	0.948	[0.45, 2]	0.888
Control / Spatial	Visualiser / Verbaliser	1.237	[0.67, 2.28]	0.497
Control / Visual	Visualiser / Verbaliser	1.548	[0.83, 2.89]	0.171
Spatial / Visual	Visualiser / Verbaliser	1.251	[0.69, 2.28]	0.465
Control / Spatial	Spatialiser / Verbaliser	0.894	[0.44, 1.82]	0.756
Control / Visual	Spatialiser / Verbaliser	1.180	[0.57, 2.43]	0.653
Spatial / Visual	Spatialiser / Verbaliser	1.320	[0.65, 2.67]	0.440

Strategy groups

```
m_acc_strat <-
  glmmTMB::glmmTMB(
    data = df_expe,
    formula = build_formula("accuracy", "strategy_group"),
    family = binomial(link = "logit"),
    prior = set_ranef_prior(55)
  )

m_acc_strat |> get_singularity()
#> The model is not singular, parameter estimates are trustworthy.

m_acc_strat |> get_performance() |> knitr::kable(align = "c")
```

AIC	BIC	R2 (cond.)	R2 (marg.)	ICC	RMSE
2922.059	3011.163	0.199	0.007	0.193	0.380

```
m_acc_strat |> report_contrast(~ strategy_group) |> knitr::kable()
#> NOTE: Results may be misleading due to involvement in interactions
```

Contrast	Odds ratio	95% CI	p.value
Visual Strategy User / No Visual Strategy	0.931	[0.66, 1.32]	0.688

```
m_acc_strat |> report_contrast(~ category | strategy_group) |> knitr::kable()
```

Contrast	strategy_group	Odds ratio	95% CI	p.value
Control / Spatial	Visual Strategy User	1.365	[0.74, 2.52]	0.457
Control / Visual	Visual Strategy User	1.574	[0.85, 2.9]	0.192
Spatial / Visual	Visual Strategy User	1.153	[0.64, 2.07]	0.837
Control / Spatial	No Visual Strategy	1.027	[0.44, 2.39]	0.997
Control / Visual	No Visual Strategy	1.362	[0.59, 3.17]	0.667
Spatial / Visual	No Visual Strategy	1.327	[0.58, 3.06]	0.707

```
m_acc_strat |> report_contrast(~ category * strategy_group, interaction = TRUE) |>
knitr::kable()
```

Category contrast	strategy_group_pairwise	Odds ratio	95% CI	p.value
Control / Spatial	Visual Strategy User / No Visual Strategy	1.330	[0.77, 2.31]	0.311
Control / Visual	Visual Strategy User / No Visual Strategy	1.156	[0.66, 2.03]	0.614
Spatial / Visual	Visual Strategy User / No Visual Strategy	0.869	[0.51, 1.49]	0.612

Visualisation

The figures showing the distribution of accuracy data displayed in the manuscript were created with the `plot_superb_jitter()` function from the package, which wraps functions from the *ggplot2* and *superb* packages to create nice visualisations easily. A little `add_significance()` helper function was also created to add significance stars to the plots.

```
library(patchwork)
library(superb)

pa1 <-
  plot_superb_jitter(
    df_expe, accuracy, group_2,
    title = "VVIQ 2 groups", y_title = "Mean accuracy",
    plot.background = ggplot2::element_rect(fill = "white", colour = NA)
  ) +
  add_significance(
    size_star = 4,
    tibble::tibble(
      x_star = 2,
      y_star = 1.07,

```

```

      stars = "*",
      x_line = .data$x_star - 0.16,
      x_line_end = .data$x_star + 0.16,
      y_line = 1.05
    )
  )

pa2 <-
  plot_superb_jitter(
    df_expe, accuracy, group_3,
    title = "VVIQ 3 groups", y_title = "Mean accuracy",
    plot.background = ggplot2::element_rect(fill = "white", colour = NA)
  ) +
  add_significance(
    size_star = 4,
    tibble::tibble(
      x_star = 3,
      y_star = 1.07,
      stars = "*",
      x_line = .data$x_star - 0.16,
      x_line_end = .data$x_star + 0.16,
      y_line = 1.05
    )
  )

pa3 <-
  plot_superb_jitter(
    df_expe, accuracy, cluster,
    title = "OSIVQ clusters", y_title = "Mean accuracy",
    plot.background = ggplot2::element_rect(fill = "white", colour = NA)
  ) +
  add_significance(
    size_star = 4,
    tibble::tibble(
      x_star = 3,
      y_star = 1.065,
      stars = "•",
      x_line = .data$x_star - 0.16,
      x_line_end = .data$x_star + 0.16,
      y_line = 1.05
    )
  )

pa <- pa1 + pa2 + pa3 +
  patchwork::plot_layout(axes = "collect", guides = "collect") &
  ggplot2::theme(legend.position = "bottom")

plot(pa)

```




All done!

```

#> - Session info -----
#> setting value
#> version R version 4.5.0 (2025-04-11 ucrt)
#> os      Windows 11 x64 (build 26100)
#> system  x86_64, mingw32
#> ui      RTerm
#> language (EN)
#> collate French_France.utf8
#> ctype   French_France.utf8
#> tz      Europe/Paris
#> date    2025-09-17
#> pandoc  3.6.3 @ C:/Program Files/RStudio/resources/app/bin/quarto/bin/tools/ (via rmarkdown)
#>
#> - Packages -----
#> ! package          * version      date (UTC) lib source
#> P abind             1.4-8        2024-09-12 [?] CRAN (R 4.5.0)
#> P aphantasiaReasoningViie * 0.0.0.9000  2025-09-16 [1] local
#> P assertthat        0.2.1        2019-03-21 [?] CRAN (R 4.5.0)
#> P boot              1.3-31       2024-08-28 [3] CRAN (R 4.5.0)
#> P class             7.3-23       2025-01-01 [3] CRAN (R 4.5.0)
#> P cli               3.6.5        2025-04-23 [?] CRAN (R 4.5.0)
#> P clue              0.3-66       2024-11-13 [?] CRAN (R 4.5.0)
#> P cluster           2.1.8.1      2025-03-12 [3] CRAN (R 4.5.0)
#> P clusterCrit       1.3.0        2023-11-23 [?] CRAN (R 4.5.0)
#> P clValid           0.7          2021-02-14 [?] CRAN (R 4.5.0)
#> P coda              0.19-4.1     2024-01-31 [?] CRAN (R 4.5.0)
#> P combinat          0.0-8        2012-10-29 [?] CRAN (R 4.5.0)
#> P crayon            1.5.3        2024-06-20 [?] CRAN (R 4.5.0)
#> P curl              6.3.0        2025-06-06 [?] RSPM
#> P diceR             3.1.0        2025-06-19 [?] RSPM (R 4.5.0)
#> P digest            0.6.37       2024-08-19 [?] CRAN (R 4.5.0)
#> P dplyr             1.1.4        2023-11-17 [?] CRAN (R 4.5.0)
#> P e1071             1.7-16       2024-09-16 [?] CRAN (R 4.5.0)
#> P emmeans           1.11.1       2025-05-04 [?] CRAN (R 4.5.0)
#> P estimability      1.5.1        2024-05-12 [?] CRAN (R 4.5.0)
#> P evaluate          1.0.4        2025-06-18 [?] RSPM (R 4.5.0)
#> P farver            2.1.2        2024-05-13 [?] CRAN (R 4.5.0)
#> P fastmap           1.2.0        2024-05-15 [?] CRAN (R 4.5.0)
#> P forcats           1.0.0        2023-01-29 [?] CRAN (R 4.5.0)
#> P foreign           0.8-90       2025-03-31 [3] CRAN (R 4.5.0)
#> P generics          0.1.4        2025-05-09 [?] CRAN (R 4.5.0)
#> P ggplot2           3.5.2        2025-04-09 [?] CRAN (R 4.5.0)
#> P glmmTMB           1.1.11       2025-04-02 [?] CRAN (R 4.5.0)
#> P glue              1.8.0        2024-09-30 [?] CRAN (R 4.5.0)
#> P gtable            0.3.6        2024-10-25 [?] CRAN (R 4.5.0)
#> P haven             2.5.5        2025-05-30 [?] RSPM (R 4.5.0)
#> P highr             0.11         2024-05-26 [?] CRAN (R 4.5.0)
#> P hms               1.1.3        2023-03-21 [?] CRAN (R 4.5.0)
#> P htmltools         0.5.8.1      2024-04-04 [?] CRAN (R 4.5.0)
#> P httpuv            1.6.16       2025-04-16 [?] CRAN (R 4.5.0)
#> P insight           1.3.1        2025-06-30 [?] RSPM (R 4.5.0)
#> P jsonlite          2.0.0        2025-03-27 [?] CRAN (R 4.5.0)
#> P klaR              1.7-3        2023-12-13 [?] CRAN (R 4.5.0)
#> P knitr             1.50         2025-03-16 [?] CRAN (R 4.5.0)

```

#> P labeling	0.4.3	2023-08-29	[?]	CRAN	(R 4.5.0)
#> P labelled	2.14.1	2025-05-06	[?]	CRAN	(R 4.5.0)
#> P later	1.4.2	2025-04-08	[?]	CRAN	(R 4.5.0)
#> P lattice	0.22-7	2025-04-02	[?]	CRAN	(R 4.5.0)
#> P lifecycle	1.0.4	2023-11-07	[?]	CRAN	(R 4.5.0)
#> P lme4	1.1-37	2025-03-26	[?]	CRAN	(R 4.5.0)
#> P lsr	0.5.2	2021-12-01	[1]	RSPM	(R 4.5.0)
#> P magrittr	2.0.3	2022-03-30	[?]	CRAN	(R 4.5.0)
#> P MASS	7.3-65	2025-02-28	[3]	CRAN	(R 4.5.0)
#> P Matrix	1.7-3	2025-03-11	[3]	CRAN	(R 4.5.0)
#> P mclust	6.1.1	2024-04-29	[?]	CRAN	(R 4.5.0)
#> P mgcv	1.9-3	2025-04-04	[?]	CRAN	(R 4.5.0)
#> P mime	0.13	2025-03-17	[?]	CRAN	(R 4.5.0)
#> P miniUI	0.1.2	2025-04-17	[?]	CRAN	(R 4.5.0)
#> P minqa	1.2.8	2024-08-17	[?]	CRAN	(R 4.5.0)
#> P mvtnorm	1.3-3	2025-01-10	[?]	CRAN	(R 4.5.0)
#> P nlme	3.1-168	2025-03-31	[3]	CRAN	(R 4.5.0)
#> P nloptr	2.2.1	2025-03-17	[?]	CRAN	(R 4.5.0)
#> P numDeriv	2016.8-1.1	2019-06-06	[?]	CRAN	(R 4.5.0)
#> P patchwork	* 1.3.1	2025-06-21	[?]	RSPM	(R 4.5.0)
#> P performance	0.14.0	2025-05-22	[?]	CRAN	(R 4.5.0)
#> P pillar	1.10.2	2025-04-05	[?]	CRAN	(R 4.5.0)
#> P pkgconfig	2.0.3	2019-09-22	[?]	CRAN	(R 4.5.0)
#> P plyr	1.8.9	2023-10-02	[?]	CRAN	(R 4.5.0)
#> P promises	1.3.3	2025-05-29	[?]	RSPM	
#> P proxy	0.4-27	2022-06-09	[?]	CRAN	(R 4.5.0)
#> P purrr	1.0.4	2025-02-05	[?]	CRAN	(R 4.5.0)
#> P questionr	0.8.1	2025-06-10	[?]	RSPM	(R 4.5.0)
#> P R6	2.6.1	2025-02-15	[?]	CRAN	(R 4.5.0)
#> P rbibutils	2.3	2024-10-04	[?]	CRAN	(R 4.5.0)
#> P RColorBrewer	1.1-3	2022-04-03	[?]	CRAN	(R 4.5.0)
#> P Rcpp	1.0.14	2025-01-12	[?]	CRAN	(R 4.5.0)
#> P Rdpack	2.6.4	2025-04-09	[?]	CRAN	(R 4.5.0)
#> P reformulas	0.4.1	2025-04-30	[?]	CRAN	(R 4.5.0)
#> P reshape2	1.4.4	2020-04-09	[?]	CRAN	(R 4.5.0)
#> P rlang	1.1.6	2025-04-11	[?]	CRAN	(R 4.5.0)
#> P rmarkdown	2.29	2024-11-04	[?]	CRAN	(R 4.5.0)
#> P rrappl	1.2.7	2024-06-26	[1]	RSPM	(R 4.5.0)
#> P rstudioapi	0.17.1	2024-10-22	[?]	CRAN	(R 4.5.0)
#> P scales	1.4.0	2025-04-24	[?]	CRAN	(R 4.5.0)
#> P sessioninfo	1.2.2	2021-12-06	[?]	RSPM	(R 4.5.0)
#> P shiny	1.10.0	2024-12-14	[?]	CRAN	(R 4.5.0)
#> P shinyBS	0.61.1	2022-04-17	[1]	RSPM	(R 4.5.0)
#> P showtext	0.9-7	2024-03-02	[?]	CRAN	(R 4.5.0)
#> P showtextdb	3.0	2020-06-04	[?]	CRAN	(R 4.5.0)
#> P stringi	1.8.7	2025-03-27	[?]	CRAN	(R 4.5.0)
#> P stringr	1.5.1	2023-11-14	[?]	CRAN	(R 4.5.0)
#> P superb	* 1.0.0	2025-08-18	[1]	RSPM	(R 4.5.0)
#> P sysfonts	0.8.9	2024-03-02	[?]	CRAN	(R 4.5.0)
#> P tibble	3.3.0	2025-06-08	[?]	RSPM	
#> P tidyr	1.3.1	2024-01-24	[?]	CRAN	(R 4.5.0)
#> P tidyselect	1.2.1	2024-03-11	[?]	CRAN	(R 4.5.0)
#> PD TMB	1.9.17	2025-03-10	[?]	CRAN	(R 4.5.0)
#> P vctrs	0.6.5	2023-12-01	[?]	CRAN	(R 4.5.0)

```

#> P withr 3.0.2 2024-10-28 [?] CRAN (R 4.5.0)
#> P xfun 0.52 2025-04-02 [?] CRAN (R 4.5.0)
#> P xtable 1.8-4 2019-04-21 [?] CRAN (R 4.5.0)
#> P yaml 2.3.10 2024-07-26 [?] CRAN (R 4.5.0)
#>
#> [1] C:/Users/Mael/AppData/Local/R/cache/R/renv/library/aphantasiaReasoningViie-d318fbb0/windows/R-4
#> [2] C:/Users/Mael/AppData/Local/R/cache/R/renv/sandbox/windows/R-4.5/x86_64-w64-mingw32/9bc714e4
#> [3] C:/Program Files/R/R-4.5.0/library
#>
#> P -- Loaded and on-disk path mismatch.
#> D -- DLL MD5 mismatch, broken installation.
#>
#> -----

```